Explaining Graph Recommendations via Counterfactual Support Sets

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Abstract. Graph-based recommender systems can provide accurate personalized recommendations by learning from user-item interaction records. However, because recommendations involve a large number of historical interactions, it is still difficult to determine which past interactions actually cause an item to appear in the recommendation list. To tackle this problem, existing explainable recommendation methods usually show similar users, related items, important neighbors, attention paths, or influential interactions. However, these methods mainly tell us which evidence looks relevant, but they do not directly answer a more important question: would the recommendation change if this evidence were removed? To address this challenge, we propose CoSRec, a method that explains graph recommendations by finding a small set of historical interactions whose removal can change the recommendation. Specifically, CoSRec keeps the trained recommender unchanged and learns which interactions of the target user should be kept or weakened. It then encourages replacing the originally recommended items with reasonable alternatives, while removing as few historical interactions as possible. The selected interactions are returned as the explanation and can be checked by recomputing the recommendation after removal. Experiments with LightGCN on MovieLens show that CoSRec can replace target recommendations with sparse edge removals while preserving most of the original recommendation list. In the all-user comparison on MovieLens, CoSRec achieves the highest Success@1, reaching 13.47% and 22.16% under sparsity budgets of 5% and 10%, respectively. The source code is available at: https://github.com/zhang-haichao/CoSRec.

Keywords: Counterfactual explanation · Graph recommendation · Explainable recommendation

1 Introduction

Personalized recommendation is a core task in modern information systems. It is widely used in e-commerce, streaming media, social platforms, and online education [1,2,7]. A recommender system helps users find useful items from a large number of choices. Recent models can produce accurate ranking results by learning from user-item interaction records. Among them, graph-based recommender systems are especially effective.

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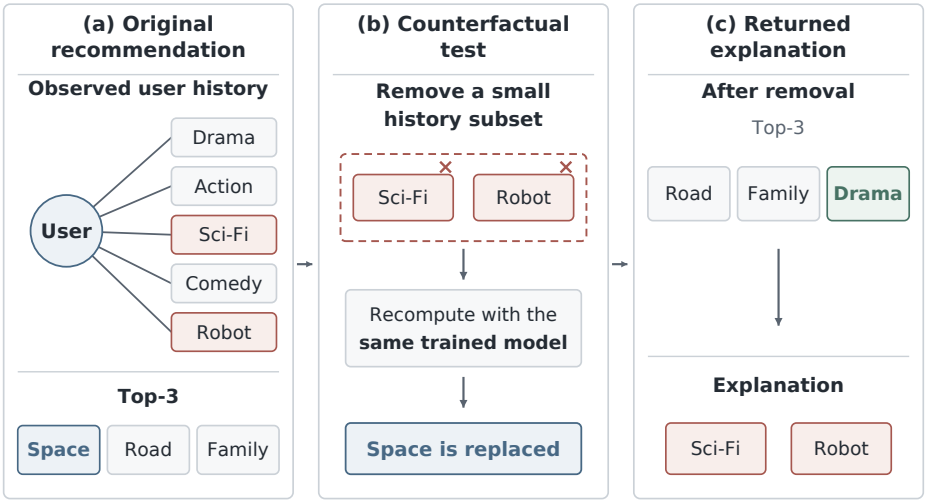


Fig. 1. Toy example of counterfactual explanation for recommendation. Starting from the original recommendation, a small subset of the user's historical interactions is removed. The recommendation is then recomputed with the same trained model. The removed interactions are returned as the explanation only if this change replaces the original recommendation.

They represent users and items as nodes, and observed interactions as edges. This structure allows the model to learn from a user's own history and from related users or items. Representative methods such as NGCF, PinSage, KGAT, and LightGCN have achieved strong recommendation performance [24,27,23,8]. Recent studies have also explored trustworthy recommendation issues such as debiasing and recommendation unlearning [34]. However, these advances do not directly explain why a particular item appears in the recommendation list. Users and developers still need to understand which historical interactions actually support a recommendation.

Many studies have explored explainable recommendations. One line of work explains recommendations by showing useful evidence, such as similar users, related items, item attributes, textual reviews, attention weights, or paths in a knowledge graph [35,6]. Another line of work studies local explanation methods for black-box models, such as LIME and SHAP [17,14]. These methods improve the transparency of recommender systems, but they still have two main limitations for graph recommendation explanation. First, many explanations are based on relevance or association. They show evidence that looks important, but they do not test whether the recommendation depends on that evidence. Second, many general explanation methods are not designed for ranking tasks. Recommendation is not a single-label prediction problem. To explain a recommendation, we need to know whether the original item can be replaced by other reasonable items in the ranked list.

Counterfactual explanation provides a direct way to address this issue [19,18,22,26]. In graph recommendation, we ask: *if a small set of the user's historical interactions had not existed, would the original item still appear in the top-K recommendation list?* If removing these interactions changes the recommendation, they can be viewed as a

counterfactual support set for the original item. As illustrated in Figure 1, this process keeps the trained recommender fixed, removes a small subset of user-history edges, and verifies the explanation by recomputing the recommendation. In this way, an interaction is selected not only because it appears relevant but also because its removal causes a concrete change in ranking. This setting is related to counterfactual recommendation studies such as CountER and Accent [18,19], but focuses specifically on graph-based recommendation and user-history edge removal [33,32,29].

To address this challenge, we propose CoSRec, a method for explaining graph recommendations through small changes to user history. CoSRec keeps the trained recommender fixed and does not update model parameters. For each target user, it learns a weight for each historical interaction. This weight indicates whether the interaction should be kept or weakened. The objective has two parts. The first part encourages the original recommended items to be replaced by reasonable alternatives. The second part limits the number of changed historical interactions. After optimization, CoSRec returns the most weakened interactions as the explanation. The explanation can then be verified by removing these interactions and recomputing the recommendation. The main contributions of this paper are summarized as follows.

- We study graph recommendation explanation from a counterfactual perspective. Instead of only showing relevant evidence, we aim to find historical interactions whose removal can change the recommendation.
- We propose CoSRec, a post-training explanation method for graph recommenders. It keeps the recommender fixed and searches for important interactions in the target user's history.
- We design a ranking-aware and sparsity-aware objective. It encourages the original recommended items to be replaced by reasonable alternatives, while keeping the explanation small and easy to inspect.
- The results show that CoSRec achieves up to 22.16% all-user Success@1 under a 10% perturbation budget, significantly outperforming baselines while maintaining high stability for the remaining recommendation list.

2 Related Work

2.1 Graph recommendation.

Graph-based recommendation treats interaction data as a bipartite graph and learns user and item representations by neighborhood aggregation. NGCF injects collaborative signals into graph propagation [24], PinSage scales graph convolution to industrial recommendation graphs [27], and KGAT integrates knowledge graph attention with recommendation [23]. LightGCN removes feature transformations and nonlinear activations, keeping only neighborhood aggregation and layer combination [8]. Surveys confirm that GNNs are now a central tool for recommendation [25]. CoSRec uses LightGCN as its main backbone because its simple propagation makes edge interventions easier to interpret.

2.2 Explainable recommendation.

Explainable recommendation aims to make personalized results transparent and useful to users [36,35,21]. Explanations may be generated from content features, user neighborhoods, textual reviews, attention weights, or knowledge graph paths. Counterfactual recommendation has recently gained attention because it provides a stronger test of model-level counterfactual dependence. CountER searches for minimal changes that reverse recommendation decisions [18], and LXR proposes a counterfactual framework for learning and evaluating recommendation explanations [4]. A RecSys 2024 study further argues that recommender performance should be considered when evaluating explainers [16].

2.3 Graph explanation and counterfactual graph learning.

GNN explanation methods identify important nodes, edges, features, or subgraphs. GNNExplainer learns masks that preserve a GNN prediction [28], PGExplainer parameterizes edge explanation across instances [15], and XGNN generates model-level graph explanations [30]. Surveys organize these methods by explanation target and evaluation setting [31,9]. Counterfactual graph explainers instead search for graph edits that change predictions. CF-GNNExplainer, robust counterfactual graph explanation, InduCE, and recent game-theoretic and unsupervised graph counterfactual explainers study this direction in classification or representation settings [13,3,20,5,10]. Recent work also warns that GNN explanations can be fragile under graph perturbations [12]. CoSRec adapts counterfactual graph explanation to personalized ranking, where the output is a top-K list, and the intervention space is the target user's historical edges.

3 Problem Formulation

Let $\mathcal{U}$ and $\mathcal{I}$ denote the sets of users and items, respectively. A recommendation graph is represented as a bipartite graph $G = (\mathcal{U} \cup \mathcal{I}, E)$, where $E \subseteq \mathcal{U} \times \mathcal{I}$ denotes the observed interaction edges. Its corresponding adjacency matrix is defined as $A \in \{0,1\}^{|\mathcal{U}|\times|\mathcal{I}|}$, where $A_{ui} = 1$ indicates that user u has interacted with item i, and 0 otherwise.

Given the adjacency matrix A and optional node features X, a graph-based recommender f_Θ parameterized by Θ maps the graph structure to d-dimensional user and item embeddings, denoted as $\mathbf{e}_u(A), \mathbf{e}_i(A) \in \mathbb{R}^d$. To capture the graph-topology dependency, the preference score for item i by user u is computed as the inner product of their representations:

$$s_{ui}(A) = \mathbf{e}_u(A)^\top \mathbf{e}_i(A). \tag{1}$$

Based on these scores, the recommender generates a ranked top-K recommendation list for user u, which is denoted by $\mathcal{R}_K(u; A)$.

For a target user u, let $\mathcal{H}_u = \{i \in \mathcal{I} \mid A_{ui} = 1\}$ be their historical interaction set. A counterfactual edge explanation aims to identify a minimal support set of historical interactions $\mathcal{S}_u \subseteq \mathcal{H}_u$ whose absence would alter the current recommendation outcome.

To formulate this intervention, we define the discrete modified adjacency matrix $A^{-\mathcal{S}_u}$ by removing the edges in $\mathcal{S}_u$:

$$(A^{-\mathcal{S}_u})_{ab} = \begin{cases} 0, & \text{if } a = u \text{ and } b \in \mathcal{S}_u, \\ A_{ab}, & \text{otherwise,} \end{cases} \tag{2}$$

where $a \in \mathcal{U}$ and $b \in \mathcal{I}$. The task of discovering the optimal counterfactual explanation is then framed as a discrete constrained optimization problem:

$$\min_{\mathcal{S}_u \subseteq \mathcal{H}_u} |\mathcal{S}_u| \quad \text{s.t.} \quad \mathcal{R}_K(u; A^{-\mathcal{S}_u}) \neq \mathcal{R}_K(u; A). \tag{3}$$

This formulation satisfies three paradigm desiderata for trustworthy model explanations: **validity** (the target recommendation must change after intervention), **sparsity** (the identified explanation set $\mathcal{S}_u$ must be compact), and **specificity** (the intervention should specifically alter the target item without broadly disrupting the rest of the user's recommendation list).

4 Method

Directly solving the discrete subset selection problem in Eq. (3) is a combinatorial optimization task that is computationally intractable for large graphs. To overcome this, we propose CoSRec, a post-hoc explainer powered by a continuous relaxation framework. As illustrated in Figure 2, the CoSRec workflow consists of three strictly aligned stages: (1) applying a *user-local retention mask* to historical interactions to generate a perturbed graph, (2) evaluating recommendation shifts through a *frozen graph recommender*, and (3) optimizing a *counterfactual ranking objective* to extract the final discrete support set.

4.1 User-Local Retention Mask and Perturbed Graph

As shown on the left side of Figure 2, the explanation process begins with the target user's historical interactions. Before any intervention, we first record the factual recommendation list $O_u = \mathcal{R}_K(u; A)$ generated by the baseline graph recommender f_Θ. The model parameters Θ are strictly frozen thereafter to ensure that any recommendation shifts are exclusively driven by graph topology changes, not model retraining.

Instead of discretely dropping edges, we assign a continuous, learnable retention variable $p_{ui} \in [0, 1]$ to each observed historical edge $(u, i) \in E$. A value of p_{ui} near 1 preserves the interaction, whereas a value near 0 attenuates it. These variables form a sparse user-local mask matrix $P^{(u)} \in [0, 1]^{|\mathcal{U}|\times|\mathcal{I}|}$:

$$P^{(u)}_{ab} = \begin{cases} p_{ui}, & \text{if } a = u \text{ and } b \in \mathcal{H}_u, \\ 1, & \text{otherwise.} \end{cases} \tag{4}$$

Using this mask, we construct the continuously relaxed *Perturbed Graph* (center of Figure 2) via element-wise Hadamard multiplication with the original adjacency matrix:

$$A'^{(u)} = A \odot P^{(u)}. \tag{5}$$

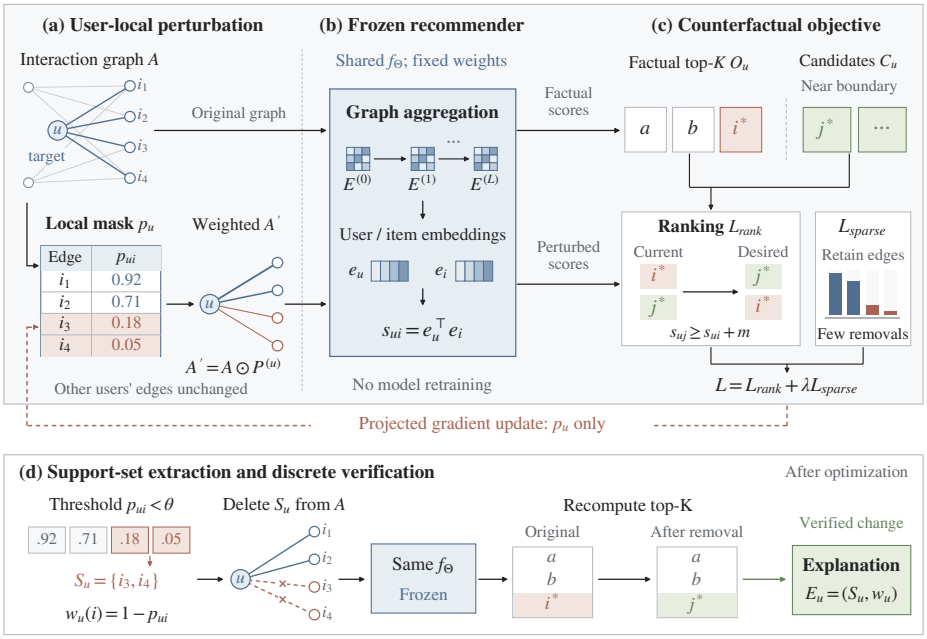


Fig. 2. Overview of CoSRec. (a) A user-local retention mask reweights the target user's historical interactions. (b) The original and perturbed graphs are evaluated by the same frozen recommender. (c) Ranking and sparsity objectives jointly optimize the mask to promote candidate items over factual recommendations with few edge removals. (d) Low-retention edges form a weighted support set, whose removal is verified to change the top-K recommendations.

This construction confines the search space to the target user's local history, leaving other users' interaction topologies unaffected.

4.2 Counterfactual Ranking Objective

Once $A'^{(u)}$ is fed into the frozen recommender, the continuous mask is optimized for two competing goals: changing the target recommendation and keeping the intervention sparse. We formulate this trade-off as:

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + \lambda \mathcal{L}_{\text{sparse}}, \tag{6}$$

where λ controls the strength of sparsity regularization. A larger λ penalizes edge suppression more strongly, leading to fewer removed edges and more stable recommendation lists.

Ranking Loss (Validity): Recommendation is fundamentally a ranking task. As visually depicted by the balance scale in Figure 2, simply reducing the preference score of a factual item is ineffective; we must simultaneously "push down" the factual items and "push up" alternative candidates.

We define a replacement candidate set $C_u \subseteq \mathcal{I} \setminus (\mathcal{H}_u \cup O_u)$ consisting of high-scoring, unobserved items near the top-K boundary. Utilizing a margin hyperparameter $m \geq 0$ and the hinge function $[x]_+ = \max(0, x)$, the ranking loss explicitly models this competitive balance:

$$\mathcal{L}_{\text{rank}} = \frac{1}{|O_u||C_u|} \sum_{i \in O_u} \sum_{j \in C_u} \left[m + s_{ui}(A'^{(u)}) - s_{uj}(A'^{(u)}) \right]_+ . \tag{7}$$

Minimizing $\mathcal{L}_{\text{rank}}$ forces the originally recommended items ($i \in O_u$) to lose their ranking advantage and be outranked by the replacement candidates ($j \in C_u$).

Sparsity Loss (Compactness): To achieve a compact and human-inspectable explanation, we penalize unnecessary edge removals. Since p_{ui} denotes the retention probability, the term $1 - p_{ui}$ represents the removal intensity. The sparsity loss calculates the average removal intensity across the user's history:

$$\mathcal{L}_{\text{sparse}} = \frac{1}{|\mathcal{H}_u|} \sum_{i \in \mathcal{H}_u} (1 - p_{ui}). \tag{8}$$

Minimizing this term drives the optimizer to suppress only a sparse fraction of critical edges, maintaining structural stability.

4.3 Optimization and Support Set Extraction

Iterative Optimization. As outlined in Algorithm 1, we initialize all retention variables to $p_{ui}^{(0)} = 1$, ensuring the optimization starts natively from the factual graph topology. At each iteration t, we compute the perturbed preference scores using $A'^{(u,t)}$ and evaluate the joint loss $\mathcal{L}$. The mask variables $\mathbf{p}_u = \{p_{ui} \mid i \in \mathcal{H}_u\}$ are then updated via projected gradient descent (PGD):

$$\mathbf{p}_u^{(t+1)} = \Pi_{[0,1]^{|\mathcal{H}_u|}} \left(\mathbf{p}_u^{(t)} - \eta \nabla_{\mathbf{p}_u} \mathcal{L} \right), \tag{9}$$

where η is the learning rate and Π projects values back into $[0, 1]$. Because the parameter cardinality is strictly bounded by the user's historical degree $|\mathcal{H}_u|$, this local optimization is highly efficient.

Extraction and Verification. Upon convergence, the continuous mask is converted back into a discrete *Counterfactual Support Set* (the final output in Figure 2) using a predefined threshold $\theta \in [0, 1]$:

$$\mathcal{S}_u = \{i \in \mathcal{H}_u \mid p_{ui} < \theta\}. \tag{10}$$

The final explanation tuple is $\mathcal{E}_u = (\mathcal{S}_u, w_u)$, where the contribution weight $w_u(i) = 1 - p_{ui}$ quantitatively ranks the counterfactual importance of each removed edge.

Finally, to bridge the continuous relaxation back to the discrete reality, we perform a strict verification step. The top-K recommendations are re-evaluated on the actual discrete graph $A^{-\mathcal{S}_u}$ to confirm that the counterfactual condition $\mathcal{R}_K(u; A^{-\mathcal{S}_u}) \neq O_u$ is flawlessly met.

Algorithm 1 CoSRec: Counterfactual Support Set Extraction

Require: Adjacency matrix A; Frozen graph recommender f_Θ; Target user u; Hyperparameters η (learning rate), λ (sparsity weights), m (margin); Max iterations T; Threshold θ.
Ensure: Counterfactual explanation tuple $\mathcal{E}_u = (\mathcal{S}_u, w_u)$.

Initialize:
Extract factual recommendation $O_u \leftarrow \mathcal{R}_K(u; A)$;
Identify historical interactions $\mathcal{H}_u \leftarrow \{i \in \mathcal{I} \mid A_{ui} = 1\}$;
Initialize retention mask: $p_{ui}^{(0)} \leftarrow 1,\ \forall i \in \mathcal{H}_u$;
Construct candidate replacement set $\mathcal{C}_u \subseteq \mathcal{I} \setminus (\mathcal{H}_u \cup O_u)$;
Optimize Continuous Mask:
for $t = 0$ **to** $T - 1$ **do**
    Construct perturbed graph $A'^{(u,t)} \leftarrow A \odot P^{(u,t)}$;
    Compute item scores via frozen f_Θ on $A'^{(u,t)}$;
    Calculate joint loss $\mathcal{L} \leftarrow \mathcal{L}_{\text{rank}} + \lambda \mathcal{L}_{\text{sparse}}$;
    Update mask via PGD: $\mathbf{p}_u^{(t+1)} \leftarrow \Pi_{[0,1]^{|\mathcal{H}_u|}} \left(\mathbf{p}_u^{(t)} - \eta \nabla_{\mathbf{p}_u} \mathcal{L} \right)$;
end for
Extract and Verify:
Extract discrete support set $\mathcal{S}_u \leftarrow \{i \in \mathcal{H}_u \mid p_{ui}^{(T)} < \theta\}$;
Calculate contribution weights $w_u(i) \leftarrow 1 - p_{ui}^{(T)},\ \forall i \in \mathcal{S}_u$;
Execute discrete verification: check if $\mathcal{R}_K(u; A^{-\mathcal{S}_u}) \neq O_u$;
return $\mathcal{E}_u = (\mathcal{S}_u, w_u)$

5 Experiments

We evaluate CoSRec from four aspects:

- **RQ1**: On sampled target cases, can CoSRec change target recommendations under sparse interventions while preserving the stability of the remaining recommendation list?
- **RQ2**: How does CoSRec perform compared with existing heuristic or counterfactual explanation methods under an all-user evaluation setting?
- **RQ3**: Can CoSRec provide concrete and verifiable user-level counterfactual explanations?
- **RQ4**: How does the sparsity penalty weight λ affect the trade-off between counterfactual success and stability?

5.1 Experiment Setup

Datasets We evaluate CoSRec on MovieLens 100K and MovieLens 1M, where ratings are binarized into user–item interactions and each dataset is split into training and test sets at an 80/20 ratio.

Baselines Since many existing counterfactual explainers are not directly designed for graph-based ranking models, direct numerical comparisons are not always compatible. To ensure a fair evaluation, we compare CoSRec with four edge-removal baselines under

Table 1. Statistics of datasets.

Dataset	Users	Items	Interactions
ML-100K	943	1,682	100,000
ML-1M	6,040	3,900	1,000,000

the same frozen LightGCN backbone. All methods operate only on the target user's historical interactions and remove the same number of edges under a given sparsity budget:

- **Random:** Randomly removes historical interactions to evaluate the graph's robustness against random noise.
- **High-Degree:** Removes historical interactions connected to the most globally popular items, testing whether recommendations are mainly driven by high-degree item popularity.
- **Attention-proxy:** Removes historical items that are most similar to the target recommendation in the embedding space, serving as a proxy for attention- or similarity-based explanations.
- **FIA:** Inspired by influence-function-based explanation [11], FIA removes historical interactions with the largest first-order influence, optionally adjusted by a diagonal second-order correction, on the target recommendation score.

Metrics We evaluate CoSRec with two metrics: Success Rate and Stability Rate. For each target setting, i.e., Top-1, Top-2, or Top-3, Success Rate measures counterfactual validity, while Stability Rate measures whether the intervention is specific to the target recommendation rather than broadly disrupting the remaining ranking list.

Success Rate. Given the counterfactual support set $\mathcal{S}_u^{(K)}$ generated for the target Top-K recommendation, Success Rate is defined as the proportion of evaluated users whose target recommendation changes after removing $\mathcal{S}_u^{(K)}$:

$$\text{Success}(K) = \frac{1}{|\mathcal{U}_{eval}|} \sum_{u \in \mathcal{U}_{eval}} \mathbf{1}\left[\mathcal{R}_K(u; A^{-\mathcal{S}_u^{(K)}}) \neq \mathcal{R}_K(u; A)\right]. \tag{11}$$

A higher Success Rate indicates stronger counterfactual validity.

Stability Rate. Stability measures the preservation of the original recommendation list after the target Top-K intervention. It is calculated as the overlap ratio of the top-N recommendation lists before and after removing the support set:

$$\text{Stability@}N = \frac{|\mathcal{R}_N(u; A) \cap \mathcal{R}_N(u; A^{-\mathcal{S}_u^{(K)}})|}{N}. \tag{12}$$

Here, K denotes the target recommendation size being replaced (which determines the counterfactual support set $\mathcal{S}_u^{(K)}$), while N represents the total length of the evaluated ranking list (we set $N = 10$ in all our experiments). A higher Stability Rate indicates better specificity, meaning the intervention alters the target item without broadly disrupting the surrounding ranking context.

Table 2. Validity-specificity trade-off of CoSRec under different sparsity budgets on sampled target cases. Target Replaced measures the percentage of sampled cases where the original target Top-K recommendation is replaced after removing the learned support set. Stability (evaluated at cutoff $N = 10$) measures how much of the overall recommendation list is preserved after the intervention.

Dataset	Target	Target Replaced (%)				Stability (%)			
		5%	10%	15%	20%	5%	10%	15%	20%
ML-100K	Top-1	83.53	85.53	86.73	87.72	91.70	91.73	90.72	90.70
	Top-2	81.73	83.53	84.63	85.63	82.56	82.34	81.68	81.43
	Top-3	72.14	72.66	76.34	77.18	74.04	73.95	73.14	73.09
ML-1M	Top-1	81.03	83.32	85.17	86.45	93.50	93.55	93.53	93.50
	Top-2	80.62	81.41	83.23	84.76	84.60	84.68	84.72	83.75
	Top-3	73.94	75.83	77.48	78.06	73.15	72.90	72.20	71.25

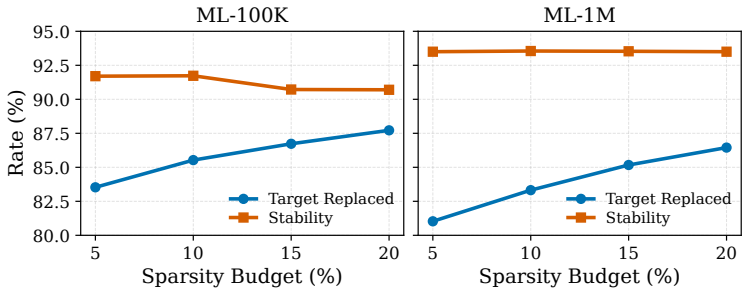


Fig. 3. Trade-off between target replacement and list preservation on sampled target cases. As the sparsity budget increases, the original Top-1 recommendation is more likely to be replaced, while the remaining recommendation list stays largely stable on both ML-100K and ML-1M.

5.2 Validity and Specificity Trade-off (RQ1)

This experiment analyzes the validity-specificity trade-off of CoSRec. To establish the upper bound of counterfactual validity, we sample N susceptible target cases where the original recommendations have near-boundary candidates. We vary the edge-removal budget from 5% to 20% on ML-100K and ML-1M, and measure whether the original target recommendation is replaced while tracking how well the remaining recommendation list is preserved. As shown in Table 2 and Figure 3, CoSRec achieves high target replacement rates under sparse interventions: with only a 5% budget, the original Top-1 recommendation is replaced in 83.53% of sampled cases on ML-100K and 81.03% on ML-1M. When the budget increases to 20%, the Top-1 replacement rate rises to 87.72% and 86.45%, respectively, while Stability remains above 90% on both datasets. These results show that, on sampled target cases, CoSRec can identify compact support sets that replace the target recommendation while preserving most of the surrounding ranking list.

5.3 Counterfactual Performance Comparison (RQ2)

Different from RQ1, which analyzes target replacement on sampled target cases, RQ2 evaluates all 943 users under a stricter all-user setting. We compare CoSRec with Random, High-Degree, Attention-proxy, and FIA, and report all-user Success@1 and Stability@10 under 5% and 10% budgets.

Table 3. Performance of counterfactual explanation methods under different sparsity budgets. The evaluation is conducted on all 943 users.

	5% Budget		**10% Budget**	
Method	Success@1 (%)↑	Stability@10 (%)↑	Success@1 (%)↑	Stability@10 (%)↑
Random	1.48	98.39	2.23	97.65
High-Degree	1.59	98.41	2.65	97.26
Attention-proxy	5.73	97.60	9.86	95.81
FIA	6.89	96.82	12.20	94.52
CoSRec (Ours)	**13.47**	95.66	**22.16**	93.15

As shown in Table 3, CoSRec consistently achieves the highest Success@1 at both budgets: under a 5% budget, it reaches 13.47%, compared with 6.89% for FIA and 5.73% for Attention-proxy; under a 10% budget, it further improves to 22.16%, clearly outperforming FIA (12.20%) and Attention-proxy (9.86%). Although CoSRec yields slightly lower Stability@10 than the weaker baselines, it still maintains high list stability at 95.66% and 93.15%, indicating that its stronger intervention effect does not come from broadly disrupting the ranking list. These results show that CoSRec provides a better trade-off between recommendation change and list preservation than existing alternatives.

5.4 Qualitative Verification of User-Level Explanations (RQ3)

This experiment examines whether CoSRec provides concrete and verifiable user-level explanations. We select representative users from MovieLens 100K and visualize each explanation as a local user–item graph, where dashed red edges denote removed support interactions and edge thickness indicates the contribution weight.

As shown in Figure 4, CoSRec changes the target recommendation by removing only a small set of historical interactions, while preserving the nearby ranking context. For User #53, removing three support edges, led by *White Squall (1996)* with $w = 0.9900$, changes the target item from *The African Queen (1951)* to *French Twist (1995)*, while the first two items in the local ranking window remain unchanged. Similarly, for User #131, the target item changes from *Bio-Dome (1996)* to *The African Queen (1951)* after removing the learned support edges. The third case further shows how CoSRec captures stronger support patterns. For User #338, all three selected support edges have high contribution weights, and removing them changes the target item from *Die Hard (1988)* to *Bean (1997)* while preserving the surrounding ranks. These cases show that

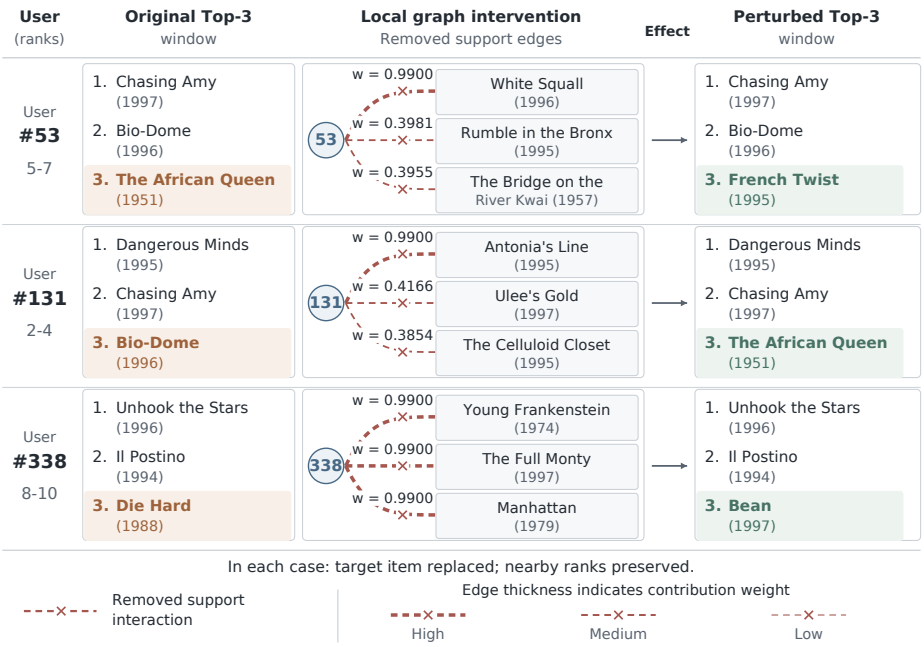


Fig. 4. Qualitative verification of CoSRec explanations on MovieLens 100K. Each row shows one user-level explanation. The original target recommendation is highlighted in orange, and the replacement item after intervention is highlighted in green. Dashed red edges with cross marks denote removed support interactions, and edge thickness indicates the learned contribution weight.

CoSRec explains recommendations through concrete historical interactions, and that the explanations can be directly verified by recomputing recommendations on the perturbed graph.

5.5 Hyperparameter Sensitivity (RQ4)

This experiment examines how the sparsity penalty weight λ affects CoSRec's counterfactual behavior. As shown in Figure 5, increasing λ makes the intervention substantially sparser: the average removed edges decrease from 6.50 to 0.40, while Success@1 drops from 28.0% to 3.0%. Meanwhile, Stability@10 increases from 0.923 to 0.996, indicating that stronger sparsity regularization better preserves the original recommendation list. These results show a clear trade-off: smaller λ improves counterfactual success through stronger perturbations, whereas larger λ favors sparse and stable explanations.

6 Conclusion

This paper studies how to explain graph recommendations from a counterfactual view. We propose CoSRec, a post-hoc method that keeps the trained recommender fixed and

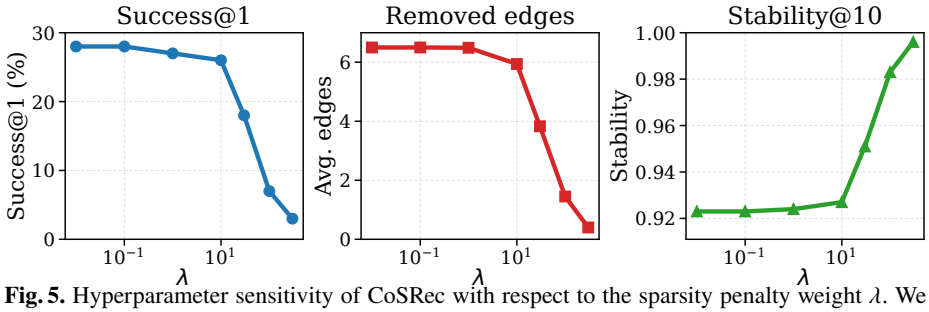


Fig. 5. Hyperparameter sensitivity of CoSRec with respect to the sparsity penalty weight λ. We report Success@1, average removed edges, and Stability@10 under different values of λ.

explains a recommendation by finding a small set of the target user's historical interactions whose removal changes the recommendation result. In this way, the explanation is not just relevant evidence, but a verifiable support set that the model actually depends on. Experiments on MovieLens show that under a strict all-user evaluation, CoSRec successfully evicts target recommendations and outperforms strong baselines with sparse edge perturbations, while preserving high stability for the rest of the recommendation list. These results show that sparse user-history intervention is a practical and effective way to explain graph recommendations, and that CoSRec provides a clear step from relevance-based explanation to counterfactual explanation.

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